

3ART12 TOUCH

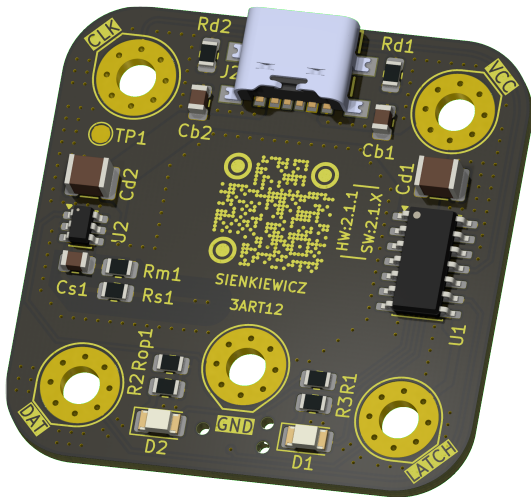
Variant: RELEASED

2026-05-21

Rev 2.1.1

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11		21		31	
2	Block Diagram	12		22		32	
3	Project Architecture	13		23		33	
4	Section A - USB connector	14		24		34	
5	Section B - uC	15		25		35	
6	Section C - Touch sensor	16		26		36	
7	Section D - Shift Register screw interface	17		27		37	
8	Requirements	18		28		38	
9	Revision History	19		29		39	
10		20		30		40	

TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

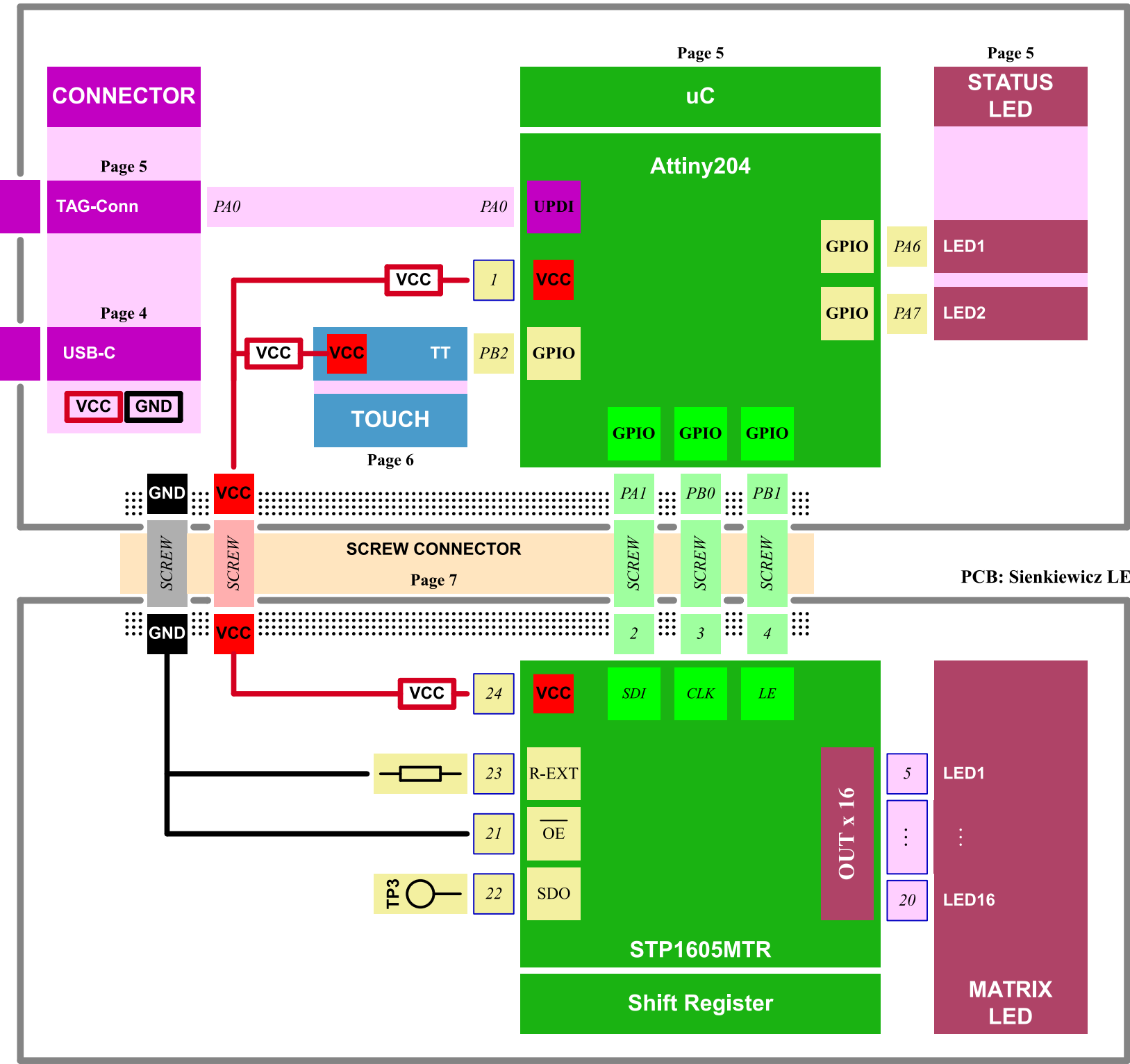
	Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: 9abb11b
		Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz	
	Sheet Title:	File Name: Sienkiewicz_TOUCH.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.1
	Sheet Path: /	Reviewer:		Size: A3	Sheet: 1 of 9

[2] Block Diagram

PCB: Sienkiewicz TOUCH

Programmer

USB-C Cable



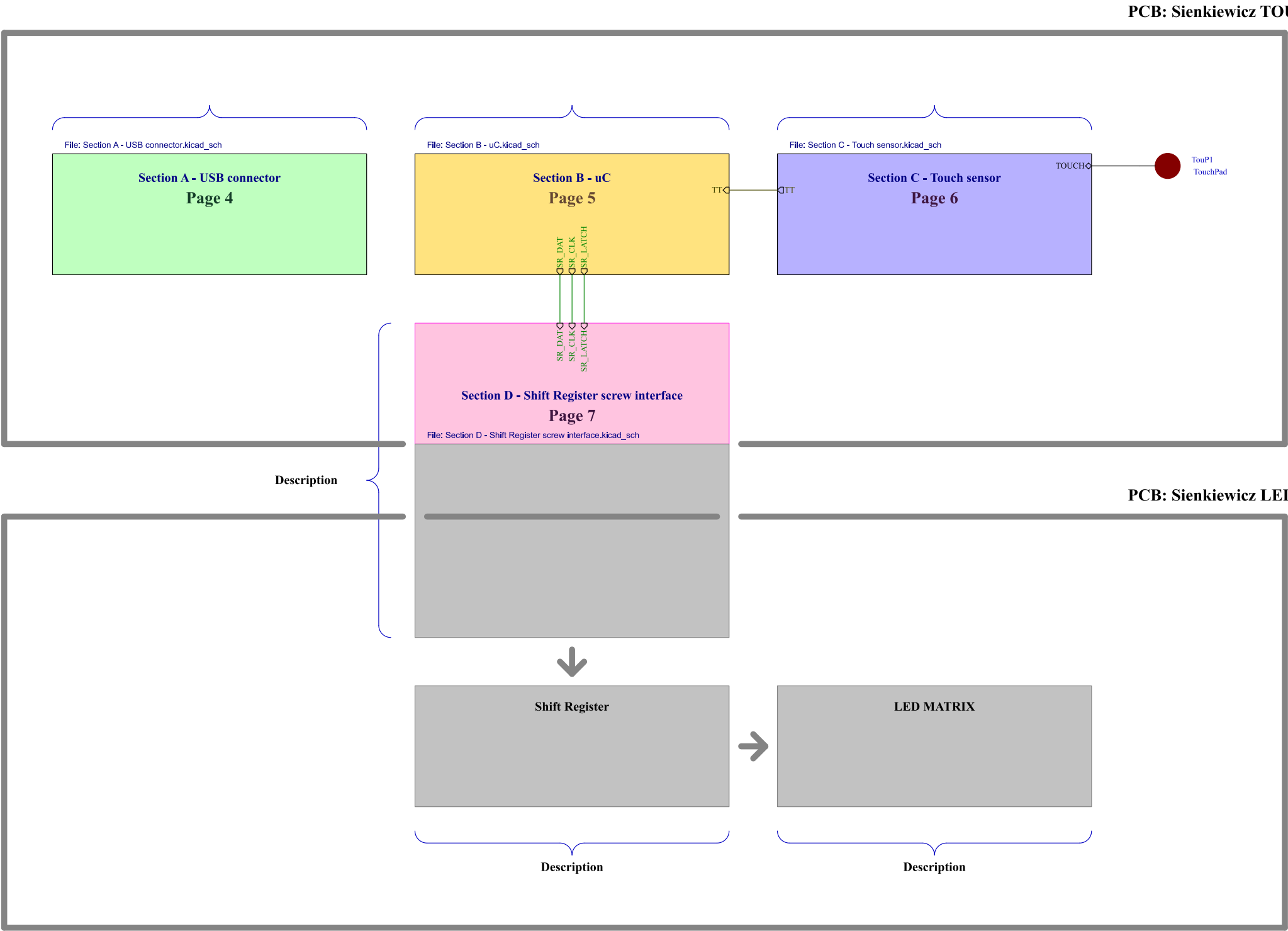
PCB: Sienkiewicz LED

Target specifications:

Input voltage	5V
Current	200mA
Spec 3	----
Spec 4	----

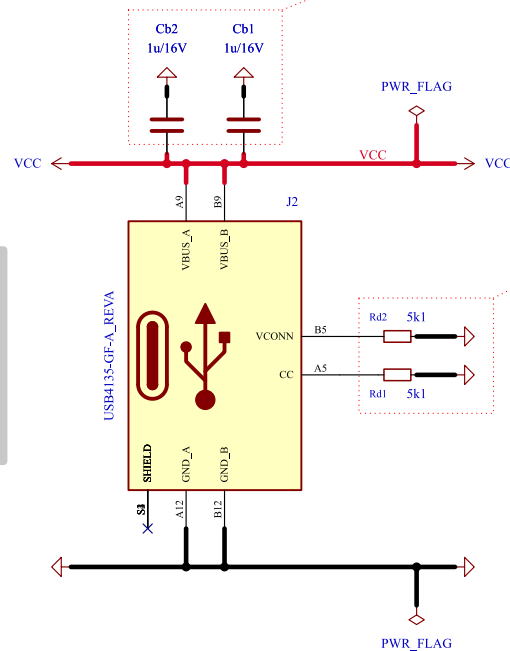
Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: 9abb11b
	Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz	
	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Revision: 2.1.1
			Sheet: 2 of 9	

[3] Project Architecture



	Comments:		Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: 9abb11b
	Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz			
	Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.1	
	Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 9	

[4] USB connector



DESIGN NOTE:
Cb1, Cb2
bulk capacitor

Release 2.0
August 2019

- 141 -

USB Type-C Cable and
Connector Specification

Table 4-2 VBUS Source Characteristics

	Minimum	Maximum	Notes
VBUS Leakage Impedance	72.4 kΩ		Leakage between VBUS pins and GND pins on receptacle when VBUS is not being sourced.
VBUS Capacitance		3000 μF	Capacitance for source-only ports between VBUS and GND pins on receptacle when VBUS is not being sourced.
		10 μF	Capacitance for DRP ports between VBUS and GND pins on receptacle when VBUS is not being sourced.

DESIGN NOTE:

Per USB-IF specification, a sink role is established by placing 5.1 kΩ pull-down (Rd) resistors on CC1 and CC2.

Release 2.0
August 2019]

- 236 -

USB Type-C Cable and
Connector Specification

Table 4-25 Sink CC Termination (Rd) Requirements

Rd Implementation	Nominal value	Can detect power capability?	Max voltage on pin
± 20% voltage clamp ¹	1.1 V	No	1.32 V
± 20% resistor to GND	5.1 kΩ	No	2.18 V
± 10% resistor to GND	5.1 kΩ	Yes	2.04 V

Notes:

1. The clamp implementation inhibits *USB PD* communication although the system can start with the clamp and transition to the resistor once it is able to do *USB PD*.

www.usb.org/sites/default/files/USB%20Type-C%20Spec%20R2.0%20-%20August%202019.pdf

Comments:

AN5346
STM32G474 Datasheet p.81
J. Pieper ADC investigation

Company:

BaJRan imgaginary industries

Variant:

RELEASED

Git Hash:

9abb11b

Board Name:

3ART12 TOUCH

Project Name:

Sienkiewicz

Sheet Title:

USB connector

File Name:

Section A - USB connector.kicad_sch

Designer:

Pawel Baran

Date:

2025-01-12

Revision:

2.1.1

Sheet Path:

/Project Architecture/Section A - USB connector/

Reviewer:

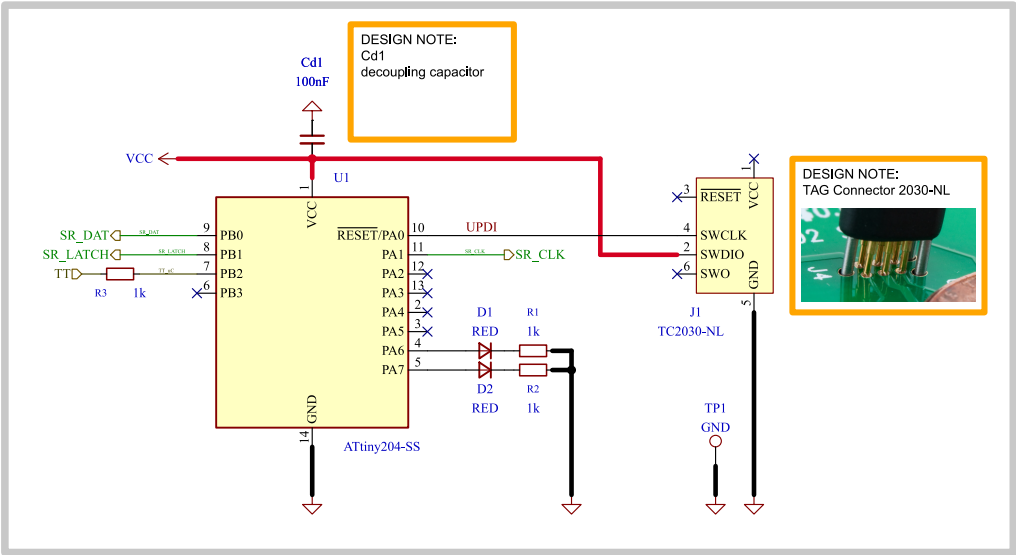
Size:

A4

Sheet:

4 of 9

[5] uC & prog connector



	Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: 9abb11b
		Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz	
	Sheet Title: uC & prog connector	File Name: Section B - uC.kicad_sch	Designer: Pawel Baran	Date: 2026-04-16	Revision: 2.1.1
	Sheet Path: /Project Architecture/Section B - uC/		Reviewer:	Size: A4	Sheet: 5 of 9

[6] Touch Sensor

DESIGN NOTE:

Table 3-2. Predefined Auto-off Delay (Active High Output)

Vt	Auto-off Delay (t _o)
Vss	Infinity (remain on until toggled to off)
Vdd	15 minutes
OUT	60 minutes

Table 3-3. Predefined Auto-off Delay (Active Low Output)

Vt	Auto-off Delay (t _o)
Vss	15 minutes
Vdd	Infinity (remain on until toggled to off)
OUT	60 minutes

DESIGN NOTE:

Output Polarity Selection

The output (OUT pin) of the QT1012 can be configured to have an active high or active low output by means of the output configuration resistor R_{op}. The resistor is connected between the output and either Vss or Vdd (see Figure 3-4 and Table 3-1). A typical value for R_{op} is 100 kΩ.

Table 3-1. Output Configuration

Name (V _{op})	Function (Output Polarity)
Vss	Active high
Vdd	Active low

DESIGN NOTE:
Cd2
decoupling
capacitor

DESIGN NOTE:
sense circuit
Cs [2nF - 60nF]
Rs [4k7 - 20k]

DESIGN NOTE:

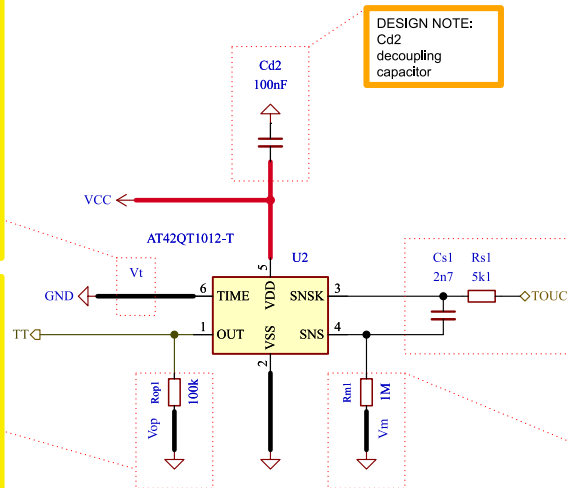
A single 1 MΩ resistor (R_m) is connected between the SNS pin and the logic level V_m to provide three auto-off functions: delay multiplication, delay override and delay retriggering. On power-up the logic level at V_m is assessed and the delay multiplication factor is set to x1 or x24 accordingly (see Figure 3-6 on page 12, Table 3-2 on page 12 and Table 3-3 on page 12). At the end of each acquisition cycle the logic level of V_m is monitored to see if an Auto-off delay override is required (see Section 3.11.5 on page 17). Setting the delay multiplier to x24 will decrease the key sensitivity. To compensate, it may be necessary to increase the value of C_s.

Table 3-4. Auto-off Delay Multiplier

V _m	Auto-off Delay Multiplier
Vss	t _o × 1
Vdd	t _o × 24

ACTIVE HIGH OUTPUT

AUTO DELAY OFF



Comments:

Company:

BaJRan imgaginary industries

Variant:

RELEASED

Git Hash:

9abb11b

Board Name:

3ART12 TOUCH

Project Name:

Sienkiewicz

Sheet Title:

Touch Sensor

File Name:

Section C - Touch sensor.kicad_sch

Designer:

Pawel Baran

Date:

2026-04-14

Revision:

2.1.1

Sheet Path:

/Project Architecture/Section C - Touch sensor/

Reviewer:

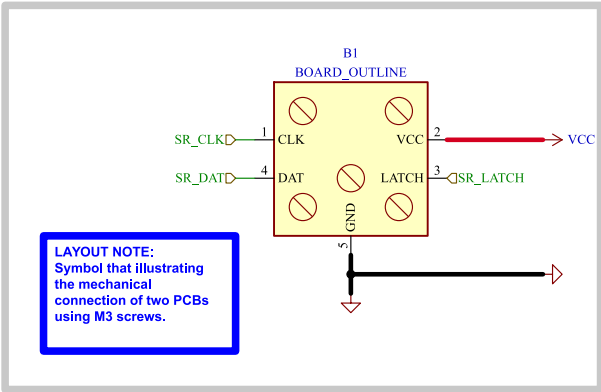
Size:

A4

Sheet:

6 of 9

[7] Shift Register screw interface



	Comments:	Company:		Variant:	Git Hash:
				RELEASED	9abb11b
		Board Name:		Project Name:	
		3ART12 TOUCH		Sienkiewicz	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Shift Register screw interface	Section D - Shift Register screw interface.kicad_sch	Pawel Baran	2026-04-16	
	Sheet Path:	Reviewer:	Size:	Sheet:	
	/Project Architecture/Section D - Shift Register screw interface/		A4	7 of 9	

[8] Requirements

Project	Simplified electronic project designed to verify practical skills in electrical engineering, PCB design, electronic assembly, embedded software implementation, and programming within the Microchip Technology development environment. Simple powering.	MEET	The project is a simple LED light effect device with a user interface realized using a capacitive touch sensor. Powered by USB-C Connector.
MAIN OBJECTIVE			

1	The project shall contain very small integrated circuits. Pitch less than 1mm	MEET	IC: Cap sensor : AT42QT1012 : SOT-23-6 package : PITCH: 0.95 IC: uC : Attiny-204 : SOIC-14_3.9x8.7mm_P1.27mm : PITCH 1.27
2	The project shall contain 0805 SMT passives components.	MEET	Device contain multiple 0805 SMT elements.
3	The project shall contain multi-pin integrated circuits to verify soldering precision.	MEET	IC: Shift register : STP16CP05MTR : SO-24 : 24 pins IC: uC : Attiny204 : SOIC-14_3.9x8.7mm : 14 pins
4	The project shall contain an UPDI programming interface for firmware upload and debugging.	MEET	uC Attiny204 with UPDI programming interface
5	Possibility to program using Microchip enviroment and pickit-4 programmer.	MEET	It is possible to program the device using pickit-4 and MPLAB software.
6	Shall contain at least two status led.	MEET	Status LED'S : LED1, LED2.
7	Shall be able to interact with user.	MEET	The user can change the light effect using a capacitive touch sensor.
8	Shall be easy to power up.	MEET	Powered by USB-C connector with general mode
9			
10			
11			
12			

		Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: 9abb11b
			Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz	
		Sheet Title: Requirements	File Name: Power - Sequencing.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.1
		Sheet Path: /Requirements/		Reviewer:	Size: A4	Sheet: 8 of 9

[9] Revision History

Version 2.0.0 - 2026-04-24

Added

- First 2.0.0 Version of pcb and schematic

Version 2.0.1 - 2026-04-25

Changed

- Modified silkscreen look, by adding bigger QR-code. small rearangement.

Version 2.0.2 - 2026-04-26

Changed

- Modified silkscreen look
- Fabrication layers update

Version 2.1.0 - 2026-05-01

Changed

- B-B connector maping changes, CLK and SDI swapped (fixes #4)

Version 2.1.1 - 2026-05-21

Changed

- B-B connector maping changes restored (fixes #4)

Added

- requirements section added for documentation

	Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: 9abb11b
		Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.1
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 9 of 9